

PAPER_859: Micro-Plasmoid 25.4 Micrometer LENR Glass Reactor Buoyancy Reversal Dynamics

Author: Daniel T. Murphy - Star Magic / UQFF Framework **Date:** 2026-04-04 **Session:** 199 **Source:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025) **Calculator:** MicroPlasmoid25umLENRBuoyancyReversalCalc (CP4 #443) **CVW:** v2.0.0 compliant

Abstract

We analyze micro-plasmoids (largest = 25.4 μm / 0.001 inch) observed in a glass reactor LENR experiment using the full quadriadic UQFF framework scaled to micrometer dimensions ($r = 25.4\text{e-}6\text{ m}$). The 165-second experiment shows 1-8 plasmoids per frame with a characteristic buoyancy reversal: initial upward swirling motion ($\sim 4\text{ }\mu\text{m/s}$) transitions to downward motion ($\sim 2\text{ }\mu\text{m/s}$) at approximately 70% of the total duration. This reversal is modeled by the F_{U_Bi} / F_{U_g} ratio exceeding unity when the Boyle's Law volume ratio $V_{\text{little}}/V_{\text{big}} = r_{\text{plasmoid}}/r_{\text{reactor}}$ amplifies buoyancy. The micro-scale LENR plasmoids represent DPM (Di-Pseudo-Monopole) creation in the early Atomic Creation Process.

1. Core Equations

- $r_{\text{plasmoid}} = 25.4\text{e-}6\text{ m}$ (0.001 inch)
 - $V_{\text{ratio}} = r_{\text{plasmoid}} / r_{\text{reactor}}$ (Boyle's Law micro-scale)
 - $F_{U_Bi} / F_{U_g} > 1 \Rightarrow$ buoyancy reversal
 - All four UQFF master equations evaluated at $r = 25.4\text{e-}6\text{ m}$, $t = 165\text{ s}$
 - $f_{Ub} = k_{Ub} * \Delta k_{\eta} * (\rho_{\text{vac},[UA]}/\rho_{\text{vac},[SCm]}) * V_{\text{ratio}}$
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2. UQFF Integration

This calculator operates as a stateless physics calculator within the Condensed-Physics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py -> bodies_*.csv data flow.

3. Source Data

- **File:** grok_share_589f6949-6fe9.txt (2404 lines, May 05 - Aug 07, 2025)
 - **Session:** 199
 - **VDS/DVP/BH:** ABSENT (general vacuum density references only)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. Srivastava, Y.N., Widom, A., Larsen, L. - Electroweak neutron production (LENR)
3. Kepler Mission DR25 - 4,034 candidates, 2,335 confirmed planets

4. Hubble Heritage Team / A. Nota (ESA/STScI) - Westerlund 2 / NGC 346 imaging
5. UQFF Calibration: $\kappa=0.0005/\text{day}$, $[\text{SSq}]=0.57$, $\beta_i \sim 0.603$